



DEPARTMENT OF THE AIR FORCE
HEADQUARTERS 72 AIR BASE WING (AFMC)
TINKER AIR FORCE BASE OKLAHOMA

29 Apr 26

MEMORANDUM FOR AFSC/PZIOC (ATTN: Sharla Brandon Flowers)
THROUGH: 72 ABW/CENM

FROM: 72 ABW/CEED
Tepa Companies

SUBJECT: Pre-Bid RFI Responses
Project: WWYK200131 - Repair Chilled Water System, B2280

Reference: Memo, 22 Apr 26, Pre-Bid RFIs, WWYK200131 - Repair Chilled Water System, B2280

1. This office reviewed the referenced RFIs for the subject project and provided answers below. Please forward to the contractor(s) for their use in preparing proposals

- (1) Please confirm the required bid bond %
Government Response: 100%.
- (2) Please confirm the amount of liquidated damages.
Government Response: \$227.75/day.
- (3) What process is being used to verify NIST SP 800-171 compliance and/or CMMC status for entities receiving access to CUI materials?
Government Response: NIST has been rescinded until further notice, but responsibility to ensure CUI safeguarding remains in prime contractors' hands. Primes can still require subs to maintain a compliant NIST score, utilize another metric to ensure safeguarding, or have subs access CUI only at prime facilities.
- (4) What controls are in place to ensure CUI is only shared with compliant prime contractors and subcontractors?
Government Response: See response to #3.
- (5) Are subcontractors required to coordinate directly with the Government to verify compliance prior to receiving CUI? If so, please confirm the process and point of contact.
Government Response: NIST has been rescinded until further notice, but responsibility to ensure CUI safeguarding remains in prime contractors' hands. See response to #3.
- (6) Will any additional guidance be issued to ensure consistent and compliant handling of CUI across all offerors?
Government Response: NIST has been rescinded until further notice, but guidance will be disseminated to contractors as it is issued.

- (7) Confirm exact piping scope: chilled water only, or any associated fittings/valves/branches tied to the chilled-water system above inaccessible ceilings and inside chases?
Government Response: Piping scope is detailed within the plans and specifications.
- (8) During light removal and reinstallation, will there be any Electrical Reinstallation testing required?
Government Response: Any lighting that was removed and reinstalled must be tested as per Tinker AFB and UFC standards.
- (9) What are the acceptable Temporary cooling and performance methods for outages?
Government Response: Phasing requirements, including temporary requirements are described on G-003.
- (10) For outage planning, does the Government anticipate one major outage, multiple phased outages, or rolling outages by area, and what maximum outage duration is acceptable for each phase?
Government Response: There is not a defined number of outages or duration of each outage that the Government is anticipating. Contractor shall provide number of outages and duration of each within their bid. Maximum outage duration will be determined at the time of the outage request based on the User's mission.
- (11) Confirm access/staging limits inside the controlled area and whether user-space swing space will be provided.
Government Response: Staging area outside the building will have to be coordinated through an airfield construction waiver. Staging areas inside the building will need to be coordinated with the user through Tepa at the time of construction.
- (12) Are there any anticipated undocumented pipes within the field or has there been a comprehensive survey of the area prior?
Government Response: Contractor shall verify existing equipment, piping, conduit, etc. prior to starting any work.
- (13) Are MICA plates required at all insulated piping segments or only specific locations? Since they appear in the submittal register.
Government Response: All insulated piping segments.
- (14) Since 23 05 93, will there be a requirement for any TAB or system recommissioning after work?
Government Response: TAB and commissioning work shall be provided as defined within Sections 01 91 00.15, 23 08 00, 23 05 93, and associated sections.
- (15) Will the Government provide additional as-built documents beyond those already referenced from B400 before award/NTP for concealed routing confirmation?
Government Response: If additional as-built drawings are required, they will need to be requested from 72 ABW/CE - Engineering Support, located in B400.
- (16) Given this project has CUI and the added time to vet applicable vendors, will the government allow additional time for technical RFI questions?
Government Response: At this time, the responsibility to ensure CUI safeguarding remains in prime contractors' hands. There are means to share CUI without compromising security or timeliness such

as restricting access to occur only at prime facilities so no additional time is currently contemplated. Albeit, if enough offerors express delays caused by vetting, proposal extensions may be considered.

- (17) Drawing ED 101, General Note 4. Please provide a list of the equipment the government would like to retain. Pricing is different from salvage vs. demolition and haul off.
 Government Response: There is no plan to have any demolished equipment returned to the Government.
- (18) Drawing ED 101, Keyed Notes 3 states, "retain existing underground concrete incased duct bank with conduits for reuse and reconnection with new work." What new work will these conduits be utilized for?
 Government Response: Refer to Sheet E-601 with new proposed OG&E XFMR to served new SWBD 5. Contractor shall omit the portion of Keyed note 3 stating to retain existing underground concrete incased duct bank for reuse and provide new as the design intent and reiterated in RFI 23 answer.
- (19) Drawing ED 401, General Note 9 states, "coordinate any "install" WOs with user before completion." Please confirm that the user will need to place a WO for CAT 6 cabling to power meter.
 Government Response: Contractor shall provide an external CAT 6 connection point within 12" center of power meter. Contractor shall also install conduit with pull string to the nearest Comm Rack (coordinate with user). When LAN drop is completed, a notification shall be sent to the COINE support team to configure Network Connection and Meter Security.
- (20) Drawing ED 401, Keyed Note 3 states to disconnect and remove existing temporary transformer (4160V-480V, 2000KVA). This transformer appears to have a rental number on it. Please confirm if the contractor is to remove transformer off site with other demolished equipment.
 Government Response: Contractor shall remove it and return it to owner for site removal to rental company. Contractor shall coordinate with Contracting Officer for exact details.
- (21) ED 601, Keyed Note 3: Contractor has reached out to OG&E, and they have no project on their books to remove OG&E transformer 347183 (4160V/2400V [Green]).
 Government Response: Contractor to coordinate with OG&E to start a work order to replace existing with new 12.5KV/72.KV-480Y/277V transformer.
- (22) E101, Keyed Note 2 states proposed OG&E transformer location. Contractor has reached out to OG&E, and they have no project on their books to add a new transformer to support a new 3000A service.
 Government Response: See previous comment (21).
- (23) OG&E transformer 347183 (4160V/2400V [Green]) currently has 2-3" concrete incased raceway entering the building. It appears the design intent is to place a new transformer (by OG&E) in the same location with 9-4" conduits to the building to feed SWBD 5. Please confirm that that is the intent.
 Government Response: The design intent is to remove and replace the secondary raceways with 9-4" conduit to feed SWBD 5.
- (24) Current underground conduits coming from OG&E transformers feeding SWGR Y-3, MCC1A, and SWGR 4160 are currently installed at the minimum depth to meet NEC 70 and reach depths in

common trench greater than 6', incased in concrete. Please confirm new underground raceways will need to be installed with a minimum 30" cover before entering the building.

Government Response: New underground raceways shall be below the frost line.

- (25) Drawing E-401, General Note 5 states, "provide additional supports to meet the project specification requirements." Please clarify the supports needed.

Government Response: This is in reference to conduit supports as required in spec section 26 20 00 and NEC 70.

- (26) Drawing E-402, Keyed Note 5: Is the heat trace on the cooling tower to be connected to the existing fire alarm system?

Government Response: No.

- (27) Drawing E-402, Keyed Note 3 states to provide housekeeping pad for SWBD 5 per 1/S-002 detail. However, S-101 does not show a housekeeping pad. Please confirm that a housekeeping pad for SWBD 5 is required.

Government Response: Equipment pad is required. Per Comment 34, there is an existing pad under the MCC-5 which can be extended to support the new SWBD 5.

- (28) Drawing E 601, Keyed Note 7 states to provide one (1) addition raceway for OG&E primary power. It has been past practice that only OG&E is allowed to install primary raceways and transformer pads. Please confirm that both the raceway and pad are not required.

Government Response: Concur OG&E will dictate if the spare conduit is needed.

- (29) Drawing E 601, Detail 2, The meter shown for SWBD 5, please confirm that it is to meet Tinker Base Standards.

Government Response: That is correct. The meter shall meet current Tinker AFB Standards.

- (30) Drawing E 601, Detail 2: Regarding the meter shown for SWBD 5, is the contract required to run CAT6 to the nearest communication room? If so, please indicate the nearest communication room located within 294".

Government Response: See previous comment (29).

- (31) Drawing E 401, Keyed Note 8, Drawing E 601, Keyed Notes 3, 5, 8: From the site visit, it appears the temporary air-cooled chillers are installed and powered. Please confirm this scope is required.

Government Response: The new temp chiller connections are only required if the contractor is going to make it work with the existing chiller equipment that is powered by the 4160V service. If they are not going to use the 4160V service, then the 480V temp chillers will be needed.

- (32) How long can the air-cooled chiller and associated equipment be down during an outage situation?

Government Response: There is not a defined number of outages or duration of each outage that the Government is anticipating. Contractor shall provide number of outages and duration of each within their bid. Maximum outage duration will be determined at the time of the outage request based on the User's mission.

- (33) Please confirm that the government will have all the Carrier cooling and associated equipment removed and it is not the contractor's responsibility.
Government Response: Temporary equipment will be removed by the Government.
- (34) There is currently a concrete pad under existing MCC-5. Please confirm that it is acceptable to extend/attach concrete to the existing pad for future SWBD 5. This will reduce the time needed during an outage.
Government Response: This is acceptable. Refer to Detail 1/S-002 for equipment pad detail.
- (35) Drawing E-401, Keyed Note 8 states to install (2) TWO 600A NEMA 3R disconnects for temporary chillers. Existing chillers are currently powered through the 4160 transformer. Please confirm these disconnects are no longer required.
Government Response: Refer to response 31.
- (36) Drawing E-401, Keyed Note 8 states to install (2) TWO 600A NEMA 3R disconnects for temporary chillers. Existing chillers are currently 1000A amps. Please confirm if the disconnects are to be increased to 1000A.
Government Response: The new disconnects are to serve the new 480V temp chiller equipment.
- (37) Most of the electrical raceway located within the west support area (Rm 123) are installed in GRC. Please confirm that EMT is allowed for new installation.
Government Response: GRC may be used for all secondary voltage wiring as per TAFB standards. EMT is allowed as per UFC, however, must meet the requirements to not be installed where subject to physical damage.
- (38) Specification section 26 24 13, Switchboards section 3.5.1.6 states to furnish certification that control systems are installed in accordance with DOD 8500.01 and DOD Instruction 8510.01. Each of these DOD publications reference how the DOD will handle CUI and nothing to do with how switchboards are to be constructed. Please provide a clear deliverable to comply with this specification section.
Government Response: This section is only for remote control or remote access capability.
Contractor shall coordinate with the Contracting Officer if required.
- (39) Specification section 26 24 13. 3.5.1.5.b(1) states to perform ground impedance measurement utilizing the fall of potential method. Given the amount of concrete in all directions around the project site, please confirm a calibrated electronic meter, such as an AEMC 3731, is acceptable in lieu of the fall of potential method.
Government Response: Contractor shall perform electrical test IAW spec section 26 24 13 §3.5.1.6(b)(1).
- (40) Please confirm that the grounding test required in specification section 26 20 00.3.5.3 is to be completed using a grounding meter such as an AEMC 3731.
Government Response: Contractor shall be IAW NETA ATS.
- (41) Specification section 26 24 13. 3.5.1.5.a.(1) states to inspect ground system for compliance with contract plans and specifications. Please confirm that the scope is limited to only project scope and not the entire grounding system.
Government Response: Confirmed the grounding test is limited to only this project scope.

- (42) Please provide current utility maps for the project area.
Government Response: See attachment 2.
- (43) Specification section 31 63 29.1.3.4 states that excavators must be licensed in the State of Oklahoma. There is not a license to operate, only training certificates. Please confirm that training certificates are acceptable.
Government Response: The training is acceptable.
- (44) How long will it take to get a Crane Permit for this project?
Government Response: The crane permit will be submitted within the Airfield Construction Waiver. See the delivery schedule in the statement of work, paragraph 3.3.3.
- (45) What is the badging process for working in this building?
Government Response: The building is located on the airfield, a controlled area. Please refer to paragraph 2.6 in the Statement of Work.
- (46) The details for the chilled water & condenser water pumps indicate vibration isolation concrete inertia bases. The specifications for chilled water equipment state vibration isolation per manufacturers' recommendations. Please advise if concrete inertia bases are required for the pumps?
Government Response: Provide concrete inertia bases for pumps.
- (47) Specification section 23 64 10, 3.5.2.c requires a 24-hour operational test of each piece of equipment for the chilled water system. Please define which pieces of chilled water system equipment you want to perform a 24-hour operational test on. Chilled water equipment on this project could include but are not limited to chillers, CHW pumps, CW pumps, VFDs, Refrigerant Monitoring System, chemical treatment system for CT, cooling towers and exhaust fans. May be redundant testing.
Government Response: One continuous 24-hour operational test for the entire chilled water system is acceptable. Operational tests for each piece of equipment within the chilled water system is not required as the single operational test will include all equipment operating together.
- (48) Will this project have the 10% hold-back in the Schedule of Values. What event triggers the release of the 10%? Is it beneficial occupancy?
Government Response: The 10% is for final acceptance. This includes the completion of every aspect of the project. All items of the project will need to be accepted (BOD), and all closeout documentation received and approved.
- (49) When will the 1-year contractor warranty start on this project? Note: depending on answer, contractor may be required to purchase additional extended warranties to cover warranty requirements.
Government Response: Warranty will start on BOD.
- (50) Specification 23 64 26 pg. 5 and 23 64 26 2.9.4 indicate requirements for testing and chemical treatment that will require submittals and approvals that will occur 12-16 months after project completion. Will the 10% hold back not be released until these requirements are met?
Government Response: Testing specified in these sections will be considered not applicable and performed by the Government. Corrective actions identified from the testing will still be the responsibility of the Contractor for the warranty period of 1 year from date of occupancy.

- (51) Will contractor warranty start be affected by those activities in question 7 above?

Government Response: Warranty will start on BOD.

- (52) Specification 23 64 26 2.5.2 indicates a Cyclonic Separator is specified. Only a side stream is shown and scheduled on the plans. Is a Cyclonic Separator required on this project?

Government Response: Cyclonic separator is not required.

- (53) Spec 23 64 26 2.5.2 requires 30% propylene glycol is required on this project.

- Is contractor responsible for replacing all the glycol in the building or just for the new work in this project?
- Can we re-use the existing glycol?
- What type of glycol is in the building currently?
- What percentage of glycol is in the current building?

Government Response:

- Response: Per General Note 1 on M-605, drain entire chilled water piping system and fill with 30% propylene glycol.
- Response: No
- Response: 30% propylene glycol system
- Response: 30% propylene glycol system

- (54) Concerning the very large existing CHW piping system in this building, we have the following questions not addressed in the RFP:

- Is there a requirement to pressure test existing CHW piping before putting thousands of dollars of glycol into the system?
- Is there a requirement to do a pre-tab on the existing CHW flows to establish a base line for the existing system or specifically AHUs?
- Is there a requirement to clean existing CHW strainers or piping?
- Is a post TAB required on the existing CHW system or AHUs?
- Who is responsible for doing closed loop treatment on the existing CHW piping system?

Government Response:

- Response: Existing and new chilled water piping shall be pressure tested per Section 23 64 26 3.4.2.
- Response: Pre-TAB of existing AHUs is required.
- Response: Existing strainers at existing AHUs served by new system shall be cleaned.
- Response: Yes, existing AHUs need to be rebalanced once new chilled water system is operating.
- Response: Chilled water system will be drained down and filled with new 30% propylene glycol under this project so contractor on this project is responsible.

- (55) Specification 23 65 00 2.7.3 requires passivation of galvanized cooling towers. This tower has SS components, does it require passivation too?

Government Response: Passivation only required for galvanized components.

- (56) Have the existing CT or condenser water piping been tested for radioactivity or hazardous materials? Who is responsible for testing and disposal of any hazardous or radioactive items on this job?

Government Response:

- (57) C102 shows CT drain connected to storm drainage; verify this is per Tinker standards.
Government Response:
- (58) The underground condenser water piping is specified to be pre-insulated PVC. Please verify insulation is required because we typically don't see this on underground CT piping at Tinker AFB.
Government Response: Insulation is not required on the below ground condenser water piping. Insulation is required on the above ground condenser water piping.
- (59) Will the Government provide an Environmental Survey?
Government Response: See attachment 1 for pictures and asbestos test results. Test results for lead-based paint could not be located.
- (60) Sheet C-102 shows five of the new piers, on the west side of the cooling towers and two of the new piers on the northeast side, outside the area of demo and re-paving. At minimum, we should have 6' x 6' openings at each pier to do our drilling and pouring, with additional repaving around the new piers. If not that, perhaps extending the demo/re-pave rectangle west and northeast past the new shafts. Please advise.
Government Response: Amount of demolition of existing paving for column construction will vary by contractor. No exception taken to additional pavement removal & replacement for column construction, however, costs for pavement removal and replacement will be included in costs of columns. No extra pay item.
- (61) Does the ductwork insulate?
Government Response: Ductwork insulation is not required.
- (62) Page M-401 Note 11 calls for heat trace on condenser water supply and return; my question is does the heat trace stop on the outer wall? Or is it going inside the building as well? And if so, do we follow specs as if it was chilled water?
Government Response: Heat trace shall be provided on exterior above ground piping per Keynote 11 on M-401 and Keynote 4 on M-402.
- (63) Will the Government evaluate an offeror's relevant past performance history (e.g. on similar scope construction projects of \$5MM or greater?) as part of the Capability and Management subfactor in Step One, or as a standalone technical subfactor?
a. It is not currently included, would the Government consider issuing an amendment to explicitly add past performance questionnaires as an evaluation criterion?
Government Response: Past performance is currently not a metric used in consideration for construction contracts. Experience is an inherent quality in displaying Capability and will underscore an offeror's approach to Technical Subfactor 1, but it will not be used in formal consideration for award.
a. N/A.
- (64) Alternatively, assuming past performance will not be a required evaluation measure; would the Government welcome (or prefer) that offerors voluntarily include a list of relevant past projects (with brief descriptions and points of contact) in their Step One technical proposal to support the evaluation

of Capability and Management and Schedule and Planning. If so, how will that information be considered?

Government Response: Offerors are not barred from submitting relevant past projects, but the Step One Technical Proposal will be evaluated on a stand-alone basis, and past performance will not be used in formal consideration for award.

- (65) Will temporary cooling/heating be required during the project?

Government Response: This project replaces an existing chilled water system so the existing heating system should not be affected. Cooling shall be maintained as described on G-003.

- (66) If temporary cooling/heating are required, please advise of the parameters required for each location within B2280 as it pertains to the temporary cooling/heating requirements.

Government Response: This project replaces an existing chilled water system so the existing heating system should not be affected. Cooling shall be maintained as described on G-003.

- (67) The SOW dated 14 Nov-25, para 2.8 Environmental Management (EM) Survey states there is LBP concerns.

a. Please provide the test locations and results.

b. Please provide detailed locations of the aforementioned LPB locations.

Government Response: See attachment 1 for pictures and asbestos test results. Test results for lead-based paint could not be located.

- (68) SOW dated 14 Nov-25, para 3.3.1 Schedule Format – refers to a reservation of the final 10% of the contract cost for final acceptance.

a. Please provide the number of days the Government has to perform its final QC Checklist after submission of the final Progress Report.

Government Response: Seven days, however the final progress report will not be accepted until all aspects of the projects have been completed.

- (69) Please provide a list of the required delegated design elements required to be performed by the subcontractor.

Government Response: Delegated design requirements are listed throughout the project documents.

- (70) The alleyway between buildings 2280 & 2283 are a mixture of concrete and asphalt surfaces. The drawings, C-102 Keynote 1, states to reinstall asphalt. Would the Government accept concrete pavement in this location?

Government Response: Yes, concrete pavement may be installed in-lieu of asphalt, provided concrete is installed per the provided concrete pavement details.

- (71) Please confirm that the alleyway between building 2280 & 2283 can be blocked off during construction.

Government Response: This cannot be confirmed and will need to go through the Airfield Construction Waiver process for approval.

- (72) Will the Government provide control points?

Government Response:

Control Point Number: 100
 Northing: 151251.06'
 Easting: 2154816.58'
 Point Elevation: 1293.77'
 Full Description: #4 BAR

Control Point Number: 101
 Northing: 151480.31'
 Easting: 2154812.06'
 Point Elevation: 1288.77'
 Full Description: #4 BAR

- (73) Will the Government provide the Geotechnical report?
 Government Response: See attachment 3.
- (74) Was a geotechnical report performed with regards to the deep foundations and/or pavement?
 Government Response: Yes, a geotechnical report was performed with regards to the deep foundation design.
- (75) Is this project tax exempt? If so, will the government provide a tax exempt form to the awarded contractor so that they may provide it to their subcontractors?
 Government Response: The project is not tax exempt.
- (76) The SF 1442, Box 13, states that the proposal validity period is 120. However, Section L, section 52.214-15, Period for Acceptance of Bids, lists 60 calendar days. Please confirm the validity period, and please confirm this validity period is for both Technical and, more importantly, Price Proposals.
 Government Response: The proposal validity period is 120 days, this stands for both Technical and Price Proposals, and Section L will be updated accordingly.
- (77) The SOW, para 2.8 lists LBP concerns. However, the drawings indicate that AFFF tanks are to be demolished. It is our understanding that the AFFF is hazardous and therefore requires specific abatement services. Outside of the LBP and the AFFF tanks, is there any other hazardous abatement required for this project?
 Government Response: The drawings indicate that the AFFF tanks will be removed under a different project, G-003, note 7. The lead based paint is the only known environmental concern.
- (78) Please confirm that no temporary services (temp air cooled chillers, electrical, water, HVAC, etc.) are required for this project?
 Government Response: Temporary cooling shall be maintained as described on G-003.
- (79) During the site visit, we observed the temp cooled chillers serving the building. The drawings don't indicate any temporary services that Contractors are to price. However, when demolition commences, how will the temp cooled chillers serve the building when we remove the pumps? It is our understanding that the temp cooled chillers serve the building pumps, which in turn serve the building. If we remove the building pumps, per the drawings, there won't be any way for the temp cooled chillers to serve the building unless temp pumps are provided by the Contractor or GOV. Please provide direction on this key concern.

Government Response: Refer to G-003 for requirements.

- (80) The structural drawings show to demo the existing concrete frame supporting the existing cooling tower. However, the new structural drawings only show concrete piers going back in its place (S-101), and what appears to be a steel frame placed on top of piers/columns though is not clear. In review of the mechanical drawings, M-502, Detail 2, there is a note stating, "Refer to structural drawings for new cooling towers steel supports. These new cooling tower steel supports are not shown on any structural drawings. Please provide details for the steel supports.

Government Response: Per the cooling tower cut sheets, the tower can be supported from piers at each anchor bolt location as an alternate support. The design intent is that the cooling tower will be anchored and supported by the piers without a steel frame.

- (81) Drawing S-101, Detail #2, shows steel lintels supporting CMU above an opening. The detail is shown on the floor plan on 1/S-101. It appears there is a new opening in the perimeter wall. However, we can't locate on the civil or structural demolition drawings information on this scope. The detail states to coordinate with mechanical drawings, but, again, we can't locate this scope. Please provide direction on this scope.

Government Response: Refer to Sheet M-401 for louver locations.

- (82) Will the Contractor be allowed to close down the alley way where this work is to be performed, to the point up to the west sidewalk adjacent to B2283? If not, it will create a significant safety hazard and concern to have non-construction personnel accessing any space east of the B2283 sidewalk.

Government Response: Please see response to question 71.

- (83) Drawing C-101, in lieu of new asphalt to patch back where the underground CW/W/CW pipes are being removed, will the GOV accept concrete?

Government Response: Yes, concrete pavement may be installed in-lieu of asphalt, provided concrete is installed per the provided concrete pavement details.

- (84) Please confirm that the existing concrete vault/monitoring well, just east of the cooling tower, is to be protected and kept in place (i.e. not demolished).

Government Response: Per callout on C-101, the existing monitoring well is to be protected in place.

- (85) Drawing S-002 includes Detail #4, Joist Reinforcement at Concentrated Load. We don't see where there is required to reinforce any existing joist. Please clarify.

Government Response: Joist reinforcement detail to be used with mechanical piping/duct work support locations.

- (86) The chiller, condenser water pumps, and cooling tower have not been identified by make, model, or capacity on the construction documents. Several notes state 'PAD DIMENSIONS ARE TO BE ADJUSTED AS NEEDED TO FIT FINAL EQUIPMENT SELECTED.' Please provide the selected equipment specifications, or confirm the date by which the government will issue a bulletin identifying

the final equipment so that piping sizes, concrete pads, electrical service, VFD sizing, and structural loading can be finalized.

Government Response: Basis of design chillers, pumps, and cooling towers can be found in the associated schedules on M-601. The final pad dimensions must be based on the associated final approved submittal that is provided by the Contractor and reviewed by the Government.

- (87) The replacement cooling tower is referenced as 'REPLACE EXISTING' but no capacity (tons), type (open/closed), weight, or footprint is specified. Please provide the cooling tower specifications, including operating and shipping weight, connection sizes, basin dimensions, and anchor bolt pattern so that roof loading, foundation, and piping can be properly designed.

Government Response: Refer to the S and M sheets and associated specifications for cooling tower and associated piping requirements.

- (88) Drawing notes state 'EXISTING TANKS TO BE REMOVED UNDER A SEPARATE PROJECT.' Please confirm:

- a. Has that separate project been awarded/scheduled?
- b. Will the tank removal be completed prior to this contractor's mobilization, or is this contractor expected to work around the existing tanks?
- c. If tank removal is concurrent, who is responsible for coordinating access and sequencing to avoid conflicts?

Government Response:

- a. The other project is in the process of being awarded.
- b. The other project will be executed prior to this project.
- c. The projects will not run concurrently.

- (89) Note on Sheet M-501 states 'PROVIDE CHILLED WATER PIPING TO OIL COOLER AS REQUIRED BY CHILLER MANUFACTURER.' Please clarify: Is oil cooler piping included in this contract's scope of work? If so, please provide the chiller manufacturer's requirements for oil cooler connection size, flow rate, and pressure drop so this work can be properly priced.

Government Response: Chilled water piping to oil cooler is only required if required by the final approved chiller. If final approved chiller submitted by the Contractor does not have an oil cooler then this requirement is void.

- (90) The civil plans show above-grade piping only, and multiple sheets reference 'REF: CIVIL PLANS FOR CONTINUATION' for the below-grade Glycol Chilled Water Supply (GCWS) and Glycol Chilled Water Return (GCWR) piping. No detail is provided showing: (a) below-grade pipe depth and routing, (b) connection point to the above-grade system, (c) pipe size and material below grade, or (d) trench protection and backfill requirements. Please provide a continuation detail for the below-grade tie-in.

Government Response: (a) Incorrect, Civil plans only shows buried water lines. Also, please reference specifications, not all information is provided in plans and details. Specification 31 00 00, 3.2.1 Water Lines states to bury water lines with minimum 3-foot cover from existing ground or proposed ground. (b) Incorrect, Keynote 3 on sheet C-102 calls out location of below ground to above ground Mechanical connections. (c) Incorrect, size of pipes is 16" and is called out with Keynote 3 on C-102. Pipe material is called out with Mechanical and is PVC. If below grade, no pipe insulation is needed, if above grade, pipe insulation is required. (d) Trench protection is part of contractor

construction means and methods and not covered in the plans and specifications. Backfill requirements are in Specification 31 00 00 Earthwork.

- (91) The equipment pad drawings indicate dimensions 'to be adjusted as needed' but provide no concrete reinforcement schedule, embedment depths for anchor bolts, or minimum pad thickness. Please provide a concrete pad detail showing: (a) pad thickness, (b) reinforcing steel size and spacing, (c) anchor bolt size, embedment depth, and edge distance, and (d) concrete compressive strength (PSI) requirement.

Government Response: Refer to Detail 1 on Sheet S-002 for equipment pad depth, reinforcement. Compressive strength to be 4,500 psi per General Notes.

- (92) The cooling tower is shown positioned on the roof/pad, but the drawings do not include details for roof penetrations, flashing, or weatherproofing for the new piping, conduit, and drainage connections. Please provide roof penetration and flashing details, including: (a) number and sizes of penetrations, (b) flashing material and method, (c) coordination with existing roofing warranty or maintenance requirements.

Government Response: As shown on numerous plans, cooling towers are located on concrete pavement not a roof.

- (93) Electrical power requirements for all major equipment (chiller, pumps, cooling tower fan motors, VFDs) cannot be finalized until equipment is selected. However, Note E-602 states 'IF CONDUCTOR RUNS EXCEED 350-FT, PROVIDE LOAD REACTORS AS REQUIRED.' Please confirm whether the VFD conductor runs from the electrical room to the equipment have been calculated to be under 350 ft, or if load reactors should be included in the bid.

Government Response: Contractor shall provide load reactors in the event the conductors are longer than 350 ft. The preliminary calculations indicate the largest length falls under 280 ft.

- (94) Sheet E-401 references a utility transformer upgrade through OG&E; Please confirm:
- Has OG&E; been notified and is the upgrade on their schedule?
 - Who is responsible for the cost of the transformer upgrade — the government, the utility, or this contractor?
 - What is the anticipated approval and installation timeline, as utility work can require 6–12 weeks?

Government Response:

- OG&E has not been notified by the EOR.
- The cost of the transformer upgrade is part of the scope of work of this project and shall be the responsibility of the contractor. Contractor shall coordinate with the Contracting Officer for further details.
- 10-14 weeks once approval is received to start work.

- (95) The drawings show the new equipment connecting to the UMCS (Utility Monitoring and Control System), but no interface protocol, point list, sequence of operations, or wiring diagram is provided. Please furnish: (a) the BMS/UMCS communication protocol (BACnet, Modbus, proprietary), (b) the

required control point list (sensor inputs, command outputs), and (c) the sequence of operations for chiller plant startup, normal operation, and shutdown.

Government Response: Refer to M-700 series plans for sequences of operation and point lists. Refer to Specification Sections 23 09 00, 23 09 13, and 23 09 23.02 for protocol requirements.

- (96) The freeze protection strategy for the outside air damper and cooling tower is referenced but not fully defined. Please provide: (a) the freezestat setpoint for damper closure, (b) whether the damper is fully closed or modulated during freeze conditions, (c) the chiller/pump interlock logic during a freeze event, and (d) glycol concentration requirement for the chilled water loop.

Government Response: Outside air damper shall operate as outlined in EXHAUST FAN CONTROL: EF-A on M-704. CHILLED WATER SYSTEM CONTROL: CHW-A on M-703 describes freeze protection. Chilled water system utilizes 30% propylene glycol per chiller and pump schedules on M-601.

- (97) The bypass valve on Sheet M-601 is shown but the operating setpoint, flow coefficient (Cv), and actuator type are not specified. Please provide: (a) differential pressure setpoint for bypass valve opening, (b) valve sizing data (Cv, body size), (c) actuator type (electric/pneumatic), and (d) fail-safe position (open/closed).

Government Response: Please clarify. M-601 is a mechanical schedules sheet and does not show a bypass valve.

- (98) The phasing plan references maintaining cooling to Building B2280 during construction, but no temporary cooling specifications, equipment requirements, or responsibility matrix is defined. Please clarify:
- Is the contractor required to provide temporary cooling, or will the government provide it?
 - If contractor-provided, what are the capacity requirements (tons)?
 - What are the connection points and power availability for temporary equipment?

Government Response:

- Refer to G-003 for temporary cooling system requirements.
- Response: Refer to G-003 for temporary cooling system requirements.
- Response: Refer to G-003 for temporary cooling system requirements.

- (99) The existing chilled water system contains refrigerant that must be recovered prior to demolition. The contract documents do not specify: (a) the refrigerant type and estimated quantity in the existing system, (b) whether refrigerant recovery and disposal is in this contractor's scope or will be handled under a separate contract, (c) the required EPA Section 608 certification for the recovery technician. Please confirm the scope and provide any known refrigerant data from the existing system.

Government Response: Contractor is responsible for determining existing refrigerant type and quantity. Existing refrigerant shall be handled as described within Specification Section 02 41 00.

- (100) The contract documents do not specify the pressure testing standard, method, or acceptance criteria for the new hydronic piping system. Please confirm: (a) the required test pressure (typically 1.5x operating pressure), (b) test duration and acceptable leakage rate, (c) test medium (water or air), (d) whether a third-party inspector or government QA representative must witness the test.

Government Response: Refer to Specification Section 23 64 26 for pressure testing requirements.

- (101) No commissioning plan, startup checklist, or performance verification criteria is included in the contract documents. Please confirm:
- Is the contractor required to provide a commissioning agent (CxA), or will the government provide commissioning?
 - What performance metrics must be demonstrated at final acceptance (e.g., chiller entering/leaving water temperatures, kW/ton efficiency, flow rates)?
 - Is a flush and chemical treatment of the hydronic system required prior to startup?
- Government Response: Commissioning, startup, and chemical treatment requirements are provided within Division 22 and 23 specification sections as well as Section 01 91 00.15. Per General Note 1 on M-605, existing chilled water system shall be drained down and filled with 30% propylene glycol.
- (102) Sheet M-101 notes that the equipment color is 'TO BE DETERMINED BY THE GOVERNMENT.' Please provide the specified color (RAL, Federal Standard, or manufacturer's standard color) at the earliest opportunity so that equipment lead times are not impacted by a late color selection.
- Government Response: M-101 does not discuss color selection at all. Where color selection is discussed, exact color will be determined by the Government during the submittal review.
- (103) The drawings do not include a piping support/hanger schedule or typical hanger detail for the new chilled water piping. Please confirm whether the contractor shall comply with MSS SP-58 hanger standards, or if a government-specific standard applies. Additionally, confirm maximum hanger spacing for the proposed pipe sizes.
- Government Response: Pipe hanger and support details are provided on M-501 and M-502. Additional pipe hanger and support information is provided with Division 22 and 23 specification sections.
- (104) Please provide the HAZMAT survey and Chain of Custody.
- Government Response: See attachment 1 for pictures and asbestos test results. Test results for lead-based paint could not be located.
- (105) Can the underground piping be performed by the mechanical subcontractor?
- Government Response: It is assumed the RFI is asking if the underground piping may be installed by the mechanical subcontractor. This will depend on if they meet the qualifications for experience and meet all conditions set forth in the project specifications. If all is in the affirmative, no exceptions taken to Mechanical subcontractor installing underground piping.
- (106) Please define temporary air equipment requirements that need to be met by the contractor.
- Government Response: Cooling shall be maintained as described in G-003.
- (107) What is the Temporary HVAC, Cooling and Heating required set points? Will this be controlled by the contractor?
- Government Response: Project replaces existing chilled water system so heating is not affected. Temporary cooling shall be maintained as described on G-003. Exact setpoints shall be coordinated with the Government during construction. Contractor to ensure temporary chilled water system remains operational throughout construction.
- (108) Please confirm that the contractor is to balance both the water and air sides.
- Government Response: All new systems including existing AHUs shall be included in the TAB scope.

- (109) Is any demolished equipment going to be returned to the government?
 Government Response: There is no plan to have any demolished equipment returned to the Government.
- (110) Schedule shows: CTM-600 duplex system, written specifications require: Steel welded tanks (Section 22 31 03, p. 4, 2.2.2). Question: Which type of system is required – CTM-600 duplex or steel welded tanks?
 Government Response: Steel welded tanks are not required. Culligan CTM-600 is the basis of design.
- (111) The drawings do not specify, however, written specifications require: Sch 80 PVC/ABS (Section 22 31 03, p. 4, 2.2.2–2.2.3.3). Question: Should the underdrain system follow the written specifications' Sch 80 PVC/ABS requirement?
 Government Response: Please clarify. Schedule 80 PVC/ABS is not described in the location referenced. Culligan CTM-600 is the basis of design so exact underdrain system can be similar to what is used in the Basis of Design.
- (112) The drawings do not specify, however, written specifications: require motor-driven piston design valve (Section 22 31 03). Question: Is the motor-driven piston valve required, or can another design be used?
 Government Response: Another design can be used. Culligan CTM-600 is the basis of design so exact valve used can be like what is used in the basis of design.
- (113) Written specs: Section 22 31 04, p. 11, 3.5.1, observation: Uncertain if required; current kits (Hach 5B) test grains per gallon. Question: Are we required to perform the written spec testing?
 Government Response: Utilizing the Hach Model 5-B Total Hardness Test Kit instead of the testing described within Section 22 31 04 3.5.1 is acceptable.
- (114) Regarding the new cooling towers:
- What is the required finished elevation for the cooling tower support system, including the design operating basin water level used for pump suction calculations?
 - What is the minimum available suction head / NPSHa the design intends to provide at each condenser pump at design and minimum-flow conditions?
- Government Response:
- Cooling tower supporting height above top of pavement is called out in Section on 4/S-101. NPSHr was determined based on top of cooling tower structural support.
 - NPSHr for each condenser water pump is provided within the PUMP SCHEDULE on M-601. Per CHILLED WATER SYSTEM CONTROL: CHW-A on M-703, condenser water pumps operate at a constant flow rate, the lowest flow through the cooling tower will occur when WCCH-4 is the only chiller operating and CWP-4 is operating utilizing one of the CTs.
- (115) The mechanical details and condenser water flow notes indicate grooved couplings are acceptable in certain locations, including tube-pull sections and vibration-isolation arrangements. Are grooved mechanical piping systems, fittings, and couplings, such as Victaulic, acceptable for above ground

chilled water and condenser water piping? Are grooved piping/fittings acceptable only at specific locations shown, or throughout the mechanical room and tower piping systems?

Government Response: Per Section 23 64 26 2.2.2.4, grooved mechanical connections may only be used in serviceable aboveground locations where temperature of circulating medium does not exceed 230°F. All above ground piping on this project is serviceable and has a circulating medium temperature below 230°F.

(116) The civil drawings show the existing 24" condenser water line to be removed and replaced with a proposed 16" condenser water line.

- a. What is the required burial depth / minimum cover?
- b. Where is the exact transition from underground to tower/pump room piping?
- c. Is buried piping to be welded / flanged / pre-insulated manufacturer-standard joints only?
- d. What are the requirements for corrosion protection, casing, trace wire, and leak detection?

Government Response:

- a. 3-ft, please see specifications.
- b. Please see Keynote 3 on sheet C-102 as well as sheet MD401 (grid E4), and sheet MD402 (grid B3).
- c. Buried PVC to be installed per manufacturer-standard installation instructions. Below grade PVC is not to be insulated, only above grade PVC to be insulated.
- d. Pipe is PVC and does not require corrosion protection. No casing is called out. See Mechanical plans/specifications for trace wire requirements. No leak detection called out with Civil plans. See Mechanical plans for additional information, sheets M-401, M-602, M-603, and M704.

(117) The phasing plan makes the contractor responsible for temporary chilled water pumps and associated piping/electrical, with pump capacity to be determined by the contractor. In order to design/size a temporary system:

- a. What is the target temporary tonnage, flow, and ΔT ?
- b. Is redundancy required?
- c. Are the existing temporary chillers to be fully functional and available?
- d. What are the required controls / monitoring during temporary operation?

Government Response:

- a. Per G-003, temporary air-cooled chillers are existing and shall remain operational during demolition and new construction phases. Exact tonnage, flow, and delta T of existing chillers shall be coordinated with the Government if selected.
- b. Per G-003, temporary air-cooled chillers are existing and shall remain operational during demolition and new construction phases. No additional temporary air-cooled chillers shall be provided.
- c. Per G-003, temporary air-cooled chillers are existing and shall remain operational during demolition and new construction phases.
- d. Per G-003, temporary air-cooled chillers are existing and shall remain operational during demolition and new construction phases.

(118) The flow notes reference chemical supplier-furnished items like the make-up meter, blowdown meter, and chemical feed connections. Is the chemical supplier also responsible for initial fill, passivation / treatment, startup chemistry, side-stream filter startup, post-start water testing, and operator training? If not, who is responsible for these tasks?

Government Response: Requirements outlined on the plans and specifications must be provided. The exact contractor who provides it is the responsibility of the General Contractor.

(119) The pump schedule references VFD use and balancing. Are the CW pumps intended to operate:

- one tower cell per pump
- common header variable flow
- staging sequence by BAS
- minimum-flow protection requirements

Government Response: Refer to CHILLED WATER SYSTEM CONTROL: CHW-A on M-703 for condenser water pump operation.

(120) The details repeatedly route low points and drain piping to floor drains. In terms of the plumbing scope of work, please confirm:

- a. Location and quantity of all required floor drains / drain receptors
- b. Capacity requirements for cooling tower blowdown, system drains, and maintenance discharge connection points for all drainage systems
- c. Whether any drainage requires neutralization or special handling (chemical treatment)

Government Response:

- a. Refer to P Series sheets for location and quantity of required floor drains/drain receptors.
- b. Drains shown on the P and C Series sheets are sized to handle drainage of chilled water equipment.
- c. When possible, chilled water drainage shall be routed back to GMU-1 per Chilled Water Flow Diagram on M-603.

(121) The structural notes require the contractor to provide equipment sizes, operating weights, vibration forces, support locations, and anchorage calculations before supports are placed. Regarding the responsibilities for structural design:

- a. Please clarify what is Engineer of Record (EOR) design vs delegated design?
- b. Is the tower support steel a fully designed (EOR) or delegated (manufacturer) design submittal?
- c. Who is responsible for anchor bolt design, vibration considerations, and equipment loading conditions?
- d. What are the submittal requirements for structural design calculations?

Government Response:

- a. EOR design are the items shown in the contract documents. Items such as equipment anchorage (embed, spacing, and anchor diameter), utility supports and connections are to be designed by Delegated Designer. Refer to General Note Sheet S-001 for additional information.
- b. Per the cooling tower cut sheets, the tower can be supported from piers at each anchor bolt location as an alternate support. The design intent is that the cooling tower will be anchored and supported by the piers without a steel frame.
- c. Anchorage and vibration considerations per delegated designer. Equipment loading on structures has been considered and accounted for in the design by the EOR.
- d. Signed and sealed calculations by an engineer registered in the State of Oklahoma. Calculations are information submittals.

(122) What are the seismic design requirements for this project?

Government Response: Refer to Design Criteria general notes on Sheet S-001.

(123) Is there a current Arc Flash / Coordination Study available?

Government Response: A preliminary study has been done in Easy Power.

(124) Are SWBD 5 and SWBD V3 entirely new gear, modifications to existing gear, or both?

Government Response: Both SWBD 5 and SWBD V3 are new and existing.

(125) It is unclear if switchboards are new or existing. Drawings indicate "new work," but they also show ties to existing gear. Please clarify.

Government Response: See comment 124.

(126) The scope involves tying into the main electrical distribution. What are the requirements for power outages and/or phasing.

Government Response: Please refer to Statement of Work 1.8 for outages.

(127) Please clarify the electrical demo scope.

- a. What feeders get removed?
- b. What stays live?
- c. What gets reused?

Government Response:

- a. Sheet ED601 indicates existing SWBD Y3 is to remain and only existing MCC-1A is to be removed but existing conduit shall remain. Existing SWBD Y1 is to remain and only MCC-5 is to be removed. Existing 4160V OG&E XFMR # 347183 (GREEN in color) shall be removed and replaced and all downstream 4160V SWGR and 4160V Chillers and pumps are to be removed.
- b. Contractor shall coordinate with mechanical contactor on phasing and refer to sheet ED601 & E-601.
- c. The existing conduit that previously served MCC-1A shall be reused, all other feeders shall not be reused.

(128) Are any upgrades to the grounding system required?

Government Response: Grounding is required for the new SWBD 5are

(129) Will the government consider/accept value engineering modifications to the scope of work? If so, should VE be broken out as options in the contractor's proposal?

Government Response: Contractor's proposals should be per the contract documents. Value engineering options will not be considered.

2. The points of contact for this project are:

- a. Cole Hixon, Project Manager. PH: 736-7085
- b. Mark Williams, Construction Inspector. PH: 582-9042
- c. Joshua Wheeler, CSAG Program Manager. PH: 582-4826
- d. Billy Deal, Contracting Officer Representative (COR). PH: 739-4975

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BILLY DEAL
Mechanical Engineer/COR
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03 Attachment(s):

1. Environmental
2. Utility Maps
3. 3. Geotechnical Report

cc:

Project folder